

NAME: _____

PERIOD: _____

Two Interfaces, Twelve Specialists

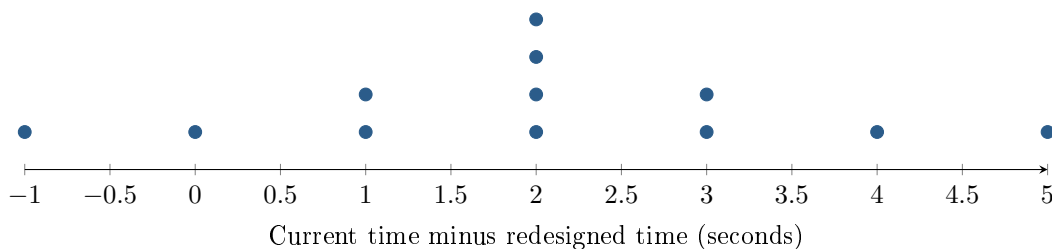
AP Statistics · Unit 4 · Topic 4.2

[STAR] THE PROBLEM

Suppose a hospital randomly samples 12 of its 180 billing specialists. Each does the same task on the current computer screen and, one week later, on a redesigned screen. Everyone uses the current screen first; there is no comparison group. Let d = current – redesigned time in seconds. The plot shows the differences; $\bar{d} = 2.000$, $s_d = 1.6514$, and the 95% critical value is $t^* = 2.201$.

Priya: “Use one sample of paired differences. Doing the tasks in a fixed order limits a cause-and-effect claim.”

Mateo: “Two columns require two independent samples. A positive interval proves the redesign caused faster work.”



FIRST TAKE — one sentence before you work the parts

Which seems more useful here: each person’s change in time or two separate groups of times? Explain in one sentence.

[BOOK] ONE-SAMPLE MEAN INTERVALS (keep on the page)

For paired measurements, define every difference in the same order and treat the n pairs as one sample: $\bar{d} \pm t^*(s_d/\sqrt{n})$ with $df = n - 1$. Check conditions on the differences: random sampling or random assignment, $n \leq 0.10N$ when sampling without replacement, and, for $n < 30$, no strong skew or outliers. Two readings from each person are not two independent samples. Random sampling supports generalization to the sampled population; random assignment or an adequate randomized comparison is needed to isolate causation from practice or time effects.

Work (a)–(c).

DOK 1 (a) How many paired differences are there, and what does a positive difference mean?

DOK 2 (b) Use the supplied summaries to calculate the 95% interval $\bar{d} \pm t^*(s_d/\sqrt{n})$. Show one calculation.

DOK 3 (c) In 3–4 sentences, choose an analysis and justify it from the linked measurements. Interpret the interval for the 180 specialists, then explain why practice or the fixed order limits a causal claim.

Frame: I would analyze _____ because each person _____.

Frame: The interval suggests _____, but _____ could also explain the change.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.